INCH-POUND

MIL-PRF-1/876F
21 September 1999
SUPERSEDING
MIL-E-1/876E
2 May 1974

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, POWER
TYPE 8165

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Tetrode.

See figure 1.

Mounting position: Vertical, base down or up.

Weight: 3-ounces (85 grams) nominal.

ABSOLUTE RATINGS: C Telegraphy

Parameter: Unit:	F1 MHz	Ef V ac	Eb V dc	Ec1 V dc	Ec2 V dc	Ib mA dc	Pp W	Pg1 W	Pg2 W	T(seal) °C	TE °C
Maximum: Minimum:	150 ---	6.3 5.7	3,000 ---	-500 ---	400 ---	150 ---	65 ---	5 ---	10 ---	200 ---	225 ---
Test conditions:	---	6.0	1,000	Adj	400	65	---	---	---	2/	---

See footnotes at end of table I.

GENERAL:

Qualifications: Required.

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TABLE I. Testing and inspection.

Inspection	Method	Notes	Conditions	Acceptance Level 3/	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 1								
Filament current	1301	---		0.65	If	3.2	3.8	A ac
Peak emission	1231	---	eb = ec1 = ec2 = 2,500 v	0.65	is	2.0	---	a
Electrode current (screen)	1256	---		0.65	Ic2	-1.0	+1.0	mA dc
Electrode voltage (grid)	1261	---		0.65	Ec1	-38.0	-58.0	V dc
Total grid current	1266	1/		0.65	Ic1	---	-5	µA dc
Primary grid emission (control)	1266	---	Ef = 6.6 V ac; Pg1 = 6 W or Ic1 = 28 mA dc; t = 15; anode and screen-grid floating	0.65	Isg1	---	-250	µA dc
Primary grid emission (screen)	1266	---	Ef = 6.6 Vac; Pg2 = 12 W or Ic2 = 27 mA dc; Ec1 = 0; t = 15; anode floating	0.65	Isg2	---	-250	µA dc
Conformance inspection, part 2								
Low-frequency vibration	1031	---	No voltages	---	---	---	---	---
Bump	1036	---	Angle = 20°	---	---	---	---	---
Bump and short	1036	---	Angle = 15°	---	---	---	---	---
Amplification factor	1316	---	g1 to g2; Ec2 = 250 V dc; Ic2 = 40 mA dc; Eb = 0	---	Mu	5.0	7.0	---
Direct-interelectrode capacitance	1331	---		---	Cgp	---	0.12	pF
					Cin	6.0	8.3	pF
					Cout	1.9	2.6	pF
Power oscillation	1236	---	Eb = 1,500 V dc; Ib = 150 mA dc; Ec2 = 250 V dc; F = 150 MHz	---	Po	110	---	W

See footnotes at end of table.

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TABLE I. Testing and inspection - Continued.

Inspection	Method	Notes	Conditions	Acceptance Level 3/	Symbol	Limits		Unit
						Min	Max	
Conformance inspection, part 3								
Life test	---	---	Group C; power oscillation; t = 500 hours	---	---	---	---	---
Life-test end points:	---	---						
Peak emission	1231	---		---	is	1.6	---	a
Primary grid emission (control)	1266	---		---	Isg1	---	-250	µA dc
Primary grid emission (screen)	1266	---		---	Isg2	---	-250	µA dc

1/ This test is to be the first test performed at the conclusion of the holding period.

2/ During all electrical tests involving application of filament power, a heat-dissipating connector (Eimac HR-6, or equivalent) may be used on the anode terminal.

3/ This specification uses accept on zero (c = 0) sampling plan, in accordance with MIL-PRF-1, Table III.

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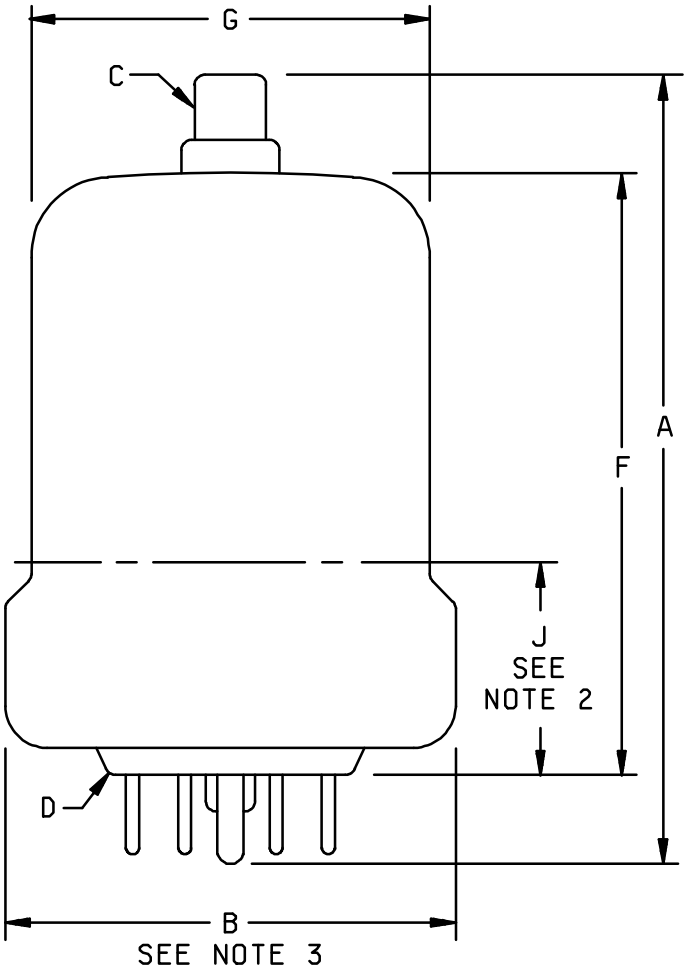

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
Conformance inspection, part 2				
A	4.000	4.375	101.60	111.13
B	---	2.375	---	60.33
G	---	2.060	---	52.32
J	.844	1.219	21.44	30.96
Conformance inspection, part 3 (see note 4)				
C	Cap: C1-22 (EIA)			
D	Base: E5-59 (See note 2) (EIA)			
F	2.938	3.313	74.63	84.15

Pin connections	
Pin No.	Element
1	f
2	g2
4	g1
6	g2
7	f
Cap	a

NOTES:

1. For pin alignment, use JEDEC gauge GE7-6. A molded glass flare base with pin configuration identical with that of base E5-59 may be used.

2. With base pins inserted into gauge GE7-6, a flat plate gauge with a 2.063 ± .003 inches (52.40 ± 0.08 mm) diameter hole is passed over diameter G until it stops at position defined by dimension J. Dimension J shall then be gauged between bottom surface of flat-plate gauge and top surface of gauge GE7-6 with surfaces of both gauges parallel. The 2.063 inches (52.40 mm) diameter hole of the flat-plate gauge and .500 inch (12.70 mm) diameter hole of gauge GE7-6 shall be concentric within .150 inch (3.81 mm).

3. A ring gauge .125 ± .010 inch (3.18 ± 0.26 mm) thick and 2.125 inches (53.98 mm) minimum and 2.128 inches (54.05 mm) maximum inside diameter shall not pass over diameter B when tried at an angle.

4. Dimensions shall be checked yearly. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of zero. In the event of failure, the test will be made as a part of conformance inspection, part 2, acceptance level 6.5, inspection level S3. The regular yearly double sampling plan may be reinstated after three consecutive samples have been accepted.

FIGURE 1. Outline drawing of electron tube type 8165.

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Air Force - 11
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Navy - AS, CG, MC, OS
Air Force - 19, 99

Preparing activity:
DLA - CC

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